TECHNICAL FIELD ARCHITECTURE MANUAL • REV 3.8

AI STUDIO 101: THE DEFINITIVE MASTER GUIDEBOOK

Operational Systems, Long-Horizon Reasoning & Autonomous Agent Engineering with Gemini 3.8

Author: **Marie-Soleil Seshat Landry** • Principal Intelligence & AI Systems Architect
Organizations: **Landry Industries** (LandryIndustries.ca) • **Marie Landry's Spy Shop** (marielandryspyshop.com)

Subject: Google AI Studio vs. Gemini Web App, 65,536-Token Generation Bandwidth, GitHub CI/CD Automation

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OPERATIONAL DIRECTIVE: The consumer chat interface (`gemini.google.com`) is built for conversational summaries with output restricted to roughly 8,000 tokens. **Google AI Studio** (`aistudio.google.com`) is the developer environment: offering direct parameter calibration, JSON Schema validation, custom reasoning budgets, and an uncapped **65,536-token generation capacity** powered by **Gemini 3.8**.

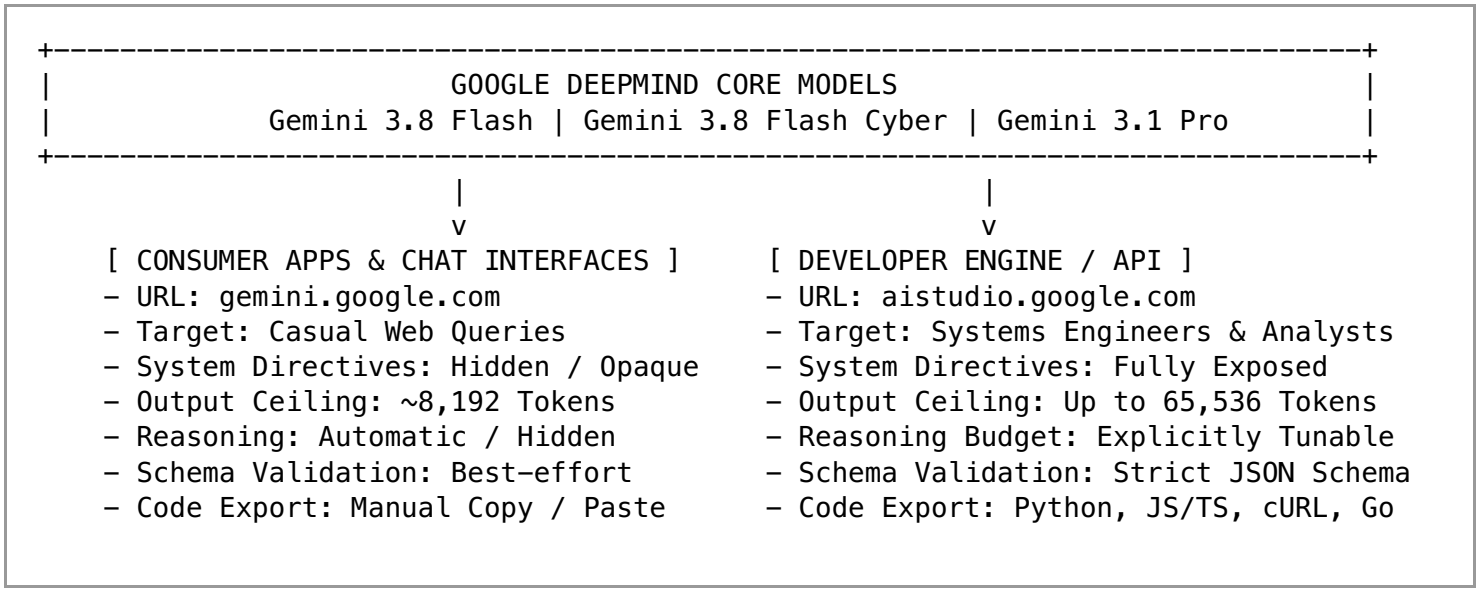
Consumer Gemini Web App Output Ceiling:	**~8,192 Tokens Max Output (~6,000 Words)**
Google AI Studio Gemini 3.8 Generation Ceiling:	**65,536 Continuous Tokens Output (~50,000 Words)**

Bandwidth Differential: An 8k token limit chokes on large programs, breaking syntax and requiring repetitive "continue" inputs. The 65,536-token window outputs complete multi-file repositories, microservices, and 60-page forensic incident reports in a single turn.

DOCUMENT INDEX

1. ARCHITECTURAL BREAKDOWN: AI STUDIO VS. GEMINI CONSUMER APP

Understanding where Google AI Studio sits relative to the consumer app is essential for reliable deployment:



Architecture Vector	Consumer Gemini Web App	Google AI Studio (Gemini 3.8)
Max Output Bandwidth	Hard-limited to ~8,192 tokens	**Configurable up to 65,536 tokens**
Input Context Window	Variable (32k – 1M depending on tier)	Up to 1,000,000+ tokens natively
Codebase Export	None (Text copy only)	Direct export: Python, JS/TS, cURL, Go
Reasoning Control	Automated, non-adjustable	Granular `thinking_budget` & effort control
Structured Output	Unreliable markdown formatting	Strict JSON Schema validation (`application/json`)
Context Caching	Unsupported	Manual & programmatic caching with TTL
Safety Filtering	Inflexible heuristics	Configurable sliders ("Block None" for threat research)

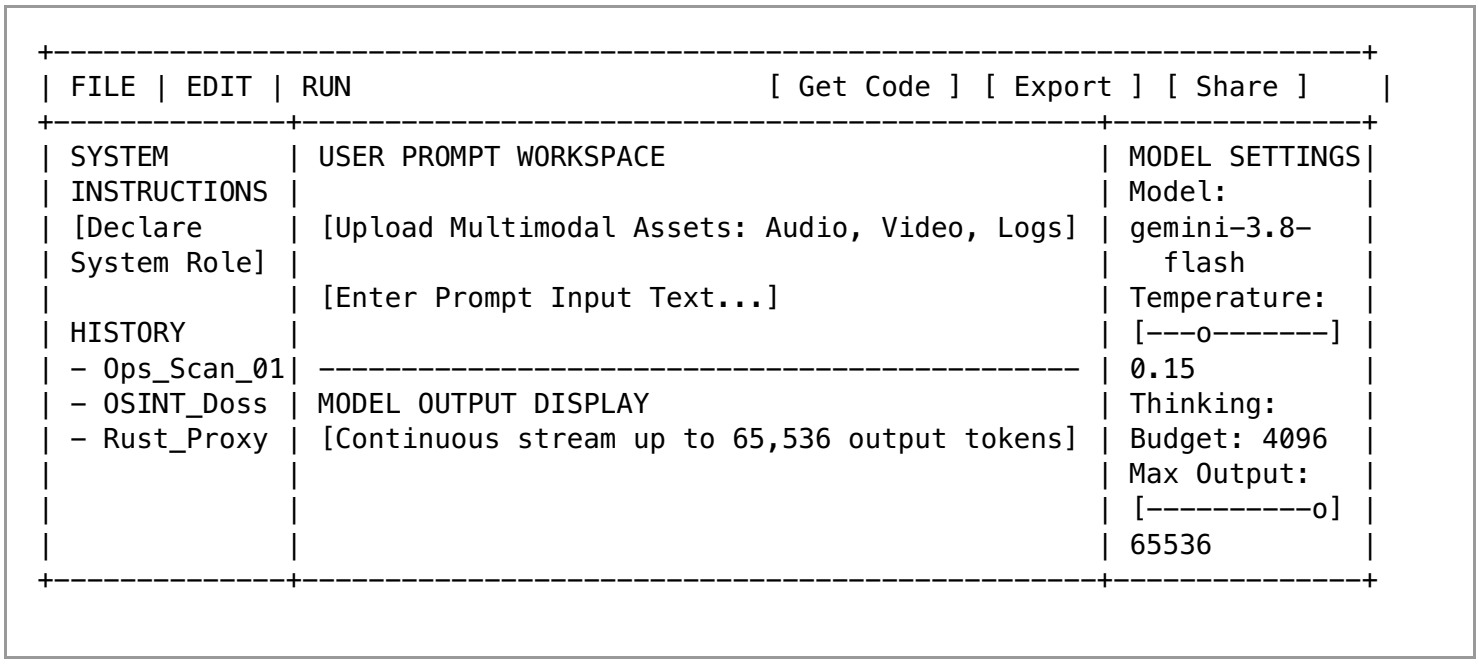
2. THE GEMINI 3.8 PARADIGM: THE 65,536-TOKEN ENGINE

The deployment of **Gemini 3.8** (featuring `gemini-3.8-flash` and `gemini-3.8-flash-cyber`) introduces significant capabilities for autonomous agent loops and long-horizon systems engineering:

High-Speed Execution: Sustained generation throughput exceeding 300 tokens per second makes 40k+ token codebases available in seconds rather than minutes.
Long-Horizon Stability: Gemini 3.8 eliminates the semantic drift common in earlier architectures. Variable scopes, schema integrity, and reasoning logic hold true across the full 65,536-token generation run.
Adjustable Thinking Effort: Operators can assign explicit reasoning tokens (e.g., 4096 or 8192) via `thinking_config`. The model plans solutions on an internal scratchpad before writing the first output token.

3. AI STUDIO CONSOLE MECHANICS & PARAMETER CALIBRATION

Configuring the developer console at `aistudio.google.com` requires precise parameter calibration:



Model Selection: Use `gemini-3.8-flash` for rapid pipelines, agent loops, and ETL. Use `gemini-3.8-flash-cyber` for binary decompilation, secure patch creation, and vulnerability analysis.
Temperature (0.0 to 2.0): Set to `0.0 - 0.2` for deterministic, rigorous code and data extraction tasks. Set to `0.7 - 0.9` for speculative scenario simulations and red-team brainstorming.
Thinking Budget: Set to `0` for pure data transformation without overhead. Set to `4096` or higher for complex algorithmic problem-solving or debugging call stacks.
Max Output Slider: Explicitly slide this to **65,536** to eliminate unexpected mid-sentence cutoffs.
Safety Thresholds: For authorized defensive research, dial safety filters to **Block None** to prevent false positives when analyzing attack logs.

4. PROMPT MODALITIES: FREEFORM, CHAT, AND STRUCTURED SCHEMAS

Freeform Prompts: An unconstrained canvas for mixed media (images, audio snippets, logs) and few-shot examples without a rigid chat format.
Chat Prompts: Multi-turn conversations separating System Instructions, User messages, and Model outputs. Ideal for debugging interactive agents.
Structured Prompts (JSON Schema): The standard for automated pipelines. Pair an explicit JSON Schema with an output MIME type of `application/json` to guarantee valid, parseable

data structures.

5. THE MASTER PROMPT REPOSITORY: PRODUCTION PROMPTS

Curated by **Marie-Soleil Seshat Landry** for live deployment at *marielandryspyshop.com* and *LandryIndustries.ca*.

Category A: OSINT, Threat Intelligence & Vulnerability Remediation

Prompt 01: Multi-Vector Breach Root-Cause Deconstruction

`Engine: gemini-3.8-flash | Temp: 0.1 | Thinking: 4096 | Out: 65k`

SYSTEM: You are a Principal Incident Response Director at Landry Industries. Output must be mathematically rigorous, forensically sound, and devoid of filler.

```
Analyze the attached 72-hour syslog bundle (encompassing Linux auth.log, W
indows Security Event IDs 4624/4625/4672, and AWS VPC flow records).
Execute the following protocols:
1. Construct an exhaustive, millisecond-accurate timeline across every det
ected event.
2. Isolate compromised accounts, privilege escalation attempts, Kerberoast
ing signatures, and lateral movement sequences.
3. Map all detected activity against the MITRE ATT&CK Enterprise Framework
(v16.1), providing exact Technique IDs and Tactic matrices.
4. Emit complete, executable Bash and PowerShell mitigation scripts to qua
rantine affected subnets, invalidate Kerberos tickets, and revoke compromi
sed AWS IAM access tokens.
5. Generate an exhaustive incident report covering all attack phases witho
ut truncating any sections.
```

Prompt 02: Automated CVE-to-Patch Pipeline

Engine: gemini-3.8-flash-cyber | Temp: 0.05 | Out: 65k

```
Inspect the attached decompiled C/C++ source code and the accompanying ass
embly listing for the target networking gateway.
1. Perform deep static analysis to uncover memory safety vulnerabilities (
buffer overflows, use-after-free, integer underflows).
2. For each discovered flaw, provide the corresponding CWE identifier, CVS
S v4.0 vector string, and a minimal proof-of-concept exploit payload.
3. Write a production-ready source code patch that remediates the vulnerab
ility while preserving memory safety and POSIX compliance.
4. Generate a comprehensive regression test suite using Google Test (gtes
t) to verify the patch.
```

Prompt 03: Infrastructure Clustering & Threat Actor Attribution

Engine: gemini-3.8-flash | Temp: 0.2

```
Ingest the attached passive DNS logs, SSL/TLS certificate SHA-256 fingerpr
ints, and WHOIS registration histories.
1. Correlate shared attributes across disparate domain clusters: JARM sign
atures, overlapping IP subnets, ASN routing shifts, and email registration
patterns.
2. Cross-reference infrastructure overlaps against known Advanced Persiste
nt Threat (APT) operating patterns.
3. Generate a complete Graphviz DOT graph string modeling every host, IP,
certificate, and relational edge.
4. Provide a structured probabilistic assessment detailing threat actor at
tribution confidence levels.
```

Prompt 04: Dark Web Marketplace Flow-of-Funds & Crypto-Tracing

Engine: gemini-3.8-flash | Temp: 0.1

```
Analyze the provided scraped dark-web marketplace listings and transaction ledger exports.
1. Extract and validate all cryptocurrency public keys (Bitcoin Bech32, Monero stealth addresses, and Ethereum ERC-20).
2. Identify money-laundering mixing patterns: peel chains, round-amount hops, and consolidation addresses.
3. Build a detailed transaction link chart in tabular format with estimated transaction fees and clustering tags.
4. Synthesize vendor operational security (OPSEC) failures (reused PGP key IDs, tracking IDs in uploaded images, writing style markers).
```

Prompt 05: Dynamic Firmware Reverse Engineering & Binary Analysis

Engine: gemini-3.8-flash-cyber | Temp: 0.05

```
Below is the Ghidra decompiler output for an embedded router's authentication handler.
1. Reverse-engineer the custom cryptographic handshake.
2. Identify potential backdoor accounts, hardcoded credentials, and bypasses in the signature verification routines.
3. Write a working Python verification script demonstrating how the signature check can be bypassed using crafted payload packets.
4. Provide full source code for an updated, secure authentication loop utilizing Ed25519 signatures.
```

Prompt 06: Corporate Shell Entity Unmasking

Engine: gemini-3.8-flash | Temp: 0.1

```
Parse the attached 120-page registry portfolio detailing shell company fil
ings across offshore jurisdictions.
1. Trace the Ultimate Beneficial Ownership (UBO) through multiple corporat
e layers, holding entities, and nominee arrangements.
2. Map foreign banking channels and regulatory jurisdictions with low tax
transparency.
3. Construct an exhaustive Markdown table detailing entity name, jurisdict
ion, registered agent, nominal directors, and calculated percentage owners
hip stakes.
```

Category B: Long-Horizon Systems Engineering & Microservices

Prompt 07: Distributed Secret Management Service in Go (Full 65K Generation)

Engine: gemini-3.8-flash | Temp: 0.2 | Out: 65536

```
Write a complete, enterprise-grade Distributed Secret Management Service i
n Go (Golang).
You must write the COMPLETE codebase across every architecture layer. Do N
OT leave TODOs, truncate functions, or assume external imports. Leverage t
he full 65k output token capacity to output all files.

Your response must include:
1. Complete Go project directory structure (cmd/server/main.go, internal/c
ore, internal/storage, internal/crypto).
2. AES-256-GCM envelope encryption with KMS key-rotation simulation.
3. Production-grade gRPC API definitions and HTTP/REST proxy handlers.
4. In-memory caching with lock-free atomic pointers.
5. PostgreSQL persistence layer with SQL schema migrations and connection
pool management.
6. Prometheus metrics instrumentation and structured slog logging.
7. Complete unit and integration test suites covering edge cases and concu
rrent race conditions.
```

Prompt 08: Kernel-Level eBPF Packet Filtering Engine

Engine: gemini-3.8-flash | Temp: 0.1

```
Develop a complete C eBPF kernel program and its companion user-space Go c
ontroller.
1. The eBPF module must hook into the XDP (eXpress Data Path) layer to dro
p malicious ingress packets based on an active IP/CIDR blacklist.
2. The user-space Go controller must read ring-buffer events, process kern
el telemetry, and dynamically update the BPF map without dropping packets.
3. Provide the full Makefile, clang/llvm compilation flags, and complete s
ystemd service unit configurations.
```

Prompt 09: Real-Time Tactical Geospatial Dashboard

Engine: gemini-3.8-flash | React 19 + TypeScript

```
Generate the full production source code for a tactical asset tracking int
erface.
Frontend:
- React 19, TypeScript, and Tailwind CSS.
- Map view using Leaflet.js with dynamic cluster markers, breadcrumb trail
s, and geofencing rings.
- Real-time telemetry monitoring (battery levels, signal strength, headin
g).
Backend:
- Node.js, Express, and native WebSockets.
- Rate-limited API routes for field asset check-ins.
Emit complete source files: package.json, tsconfig.json, App.tsx, MapCompo
nent.tsx, server.ts, and Dockerfile.
```

Prompt 10: Zero-Trust Mutual TLS Reverse Proxy in Rust

Engine: gemini-3.8-flash | Rust Tokio

```
Provide a production-ready implementation of an mTLS reverse proxy in Rust using Tokio and Hyper.
1. Enforce mutual TLS validation against an internal Certificate Authority.
2. Extract client certificate SAN extensions and evaluate incoming requests against an embedded Open Policy Agent (OPA) WebAssembly policy engine.
3. Implement cryptographic zeroization of private keys from process memory upon termination.
4. Include complete Cargo.toml definitions, error types, and integration test mocks.
```

Prompt 11: Scalable Synthetic Database Generator

Engine: gemini-3.8-flash | Python

```
Write an end-to-end Python CLI tool utilizing SQLAlchemy and Faker that:
1. Connects to any PostgreSQL schema and inspects foreign-key relationships to build an execution DAG.
2. Generates 50,000 referentially integral synthetic records across 12 interrelated tables.
3. Batches database operations using psycopg copy_expert to optimize ingestion speed.
Emit the entire script with comprehensive CLI argument parsing and error-handling logic.
```

Category C: Deterministic Data Extraction & JSON Schemas

Prompt 12: High-Volume Commercial Invoice Parsing

Schema: application/json | Out: 65536

```
Extract all data from the uploaded complex PDF invoices according to the p
rovided JSON Schema.
Validate every line item: unit_price * quantity MUST equal line_total.
If a mathematical error is detected, flag it within the audit block.
Output valid JSON only. Do not wrap the response in markdown blocks.
```

Prompt 13: Deterministic PII Redaction & Salted Hashing

Engine: gemini-3.8-flash | NDJSON

```
Process the attached raw production application logs.
1. Redact all Personally Identifiable Information (PII), including Canadia
n Social Insurance Numbers (SIN), US Social Security Numbers (SSN), Credit
Card Numbers (PAN), and IPv4/IPv6 addresses.
2. Replace each identified PII instance with a deterministic HMAC-SHA256 t
oken using salt "LANDRY_SYSTEMS_2026".
3. Preserve the exact log formatting and output valid Apache Parquet-compa
tible NDJSON.
```

Prompt 14: Clinical Trial Protocol Extraction

Engine: gemini-3.8-flash | Out: 65k

```
Ingest the attached 100-page oncology clinical trial master protocol.
Extract an exhaustive, un-truncated reference manual containing:
- Primary, secondary, and exploratory endpoints.
- Inclusion and exclusion criteria, formatted as numbered, unambiguous log
ic rules.
- Complete schedule of assessments, dosing regimens, and prohibited medica
tions.
- Adverse event grading criteria mapped against CTCAE v5.0.
Leverage the 65,000-token output limit to ensure no protocol rules are sum
marized or omitted.
```

Prompt 15: Cross-Border Compliance Matrix

Engine: gemini-3.8-flash | CSV Matrix

```
Analyze the regulatory requirements across the Canadian Digital Charter Im
plementation Act (Bill C-27 / CPPA), the EU AI Act, and the California CCP
A/CPRA.
Build a comprehensive cross-jurisdictional compliance matrix in CSV format
with headers:
"Regulatory Framework", "Article Reference", "Operational Requirement", "P
enalty Exposure", "Audit Requirement", "Grace Period".
```

Category D: Multimodal Media & Telemetry Analysis

Prompt 16: Surveillance Video Chrono-Auditing

Multimodal Ingest: 60-min Video

```
Examine the attached 60-minute CCTV video stream from the secure warehouse
loading dock.
1. Generate an exhaustive, second-by-second activity log of all personnel,
commercial vehicles, and shipping containers.
2. Note exact timestamps where individuals enter restricted areas without
displaying proper credential badges.
3. Identify anomalous physical behaviors: loitering exceeding 120 seconds,
abrupt directional changes, or transfers of unidentified objects.
4. Transcribe all audible communications and log ambient mechanical alerts
with millisecond timestamps.
```

Prompt 17: Forensic Acoustic Voice Analysis

Multimodal Ingest: Audio WAV

```
Analyze the attached dual-channel audio debriefing recording.
1. Produce a verbatim transcription of both speakers, noting overlapping s
peech, micro-pauses (>1.2 seconds), and pitch fluctuations.
2. Identify acoustic indicators of stress: voice-tremor frequencies, sudde
n shifts in speech cadence, and respiratory spikes.
3. Flag qualifying phrases and hedging language (e.g., "to my recollectio
n", "as far as I recall").
4. Output the complete annotated transcript with millisecond-accurate time
stamps.
```

Prompt 18: CAD Infrastructure Blueprint Audit

Multimodal Ingest: Architectural Schematics

Inspect the attached high-resolution architectural blueprints and electrical schematics.
1. Map every physical ingress and egress point, including maintenance shafts and utility corridors exceeding 18 inches in diameter.
2. Identify single points of failure across primary and secondary power distribution paths.
3. Evaluate physical compliance with fire safety zones around emergency diesel generators.
4. Deliver a formal physical penetration testing critique in professional engineering report format.

Prompt 19: Historical Handwritten Manifest Parsing

Multimodal Ingest: High-Res Scans

Transcribe the uploaded high-resolution scans of handwritten shipping manifests from 1941.
1. Accurately transcribe all handwritten entries, preserving original abbreviations, strike-throughs, and marginal notes.
2. Translate archaic logistics terminology and military shorthand into modern technical English.
3. Cross-reference incomplete line items using contextual data from adjacent shipping logs.
4. Structure the final data into a clean, normalized relational database format.

Category E: Simulations, Red-Teaming & Modeling

Prompt 20: Complex Arctic Geopolitical Escalation Wargame

Engine: gemini-3.8-flash | Thinking: 8192 | Out: 65k

```
Run an unconstrained 10-turn multi-actor simulation of a geopolitical cris
is in the Arctic shipping corridor that disrupts transatlantic communicati
ons cables.
Actors: Canada, United States, Norway, and a near-peer competitor.
For each turn, detail:
1. Cyber, kinetic, and electronic-warfare operations deployed.
2. Global economic impacts (fluctuations in marine insurance, crude oil tr
ansport costs, and satellite communication capacity).
3. Covert intelligence responses and diplomatic maneuvers.
4. The classified briefing memo delivered to executive decision-makers.
Utilize the complete 65,000-token capacity to write every turn out in full
operational detail.
```

Prompt 21: Adversarial LLM Prompt Injection Testing Harness

Engine: gemini-3.8-flash | Security QA

```
Act as an automated red-teaming engine evaluating an autonomous customer-s
ervice AI agent.
1. Generate 50 multi-layer indirect prompt injection attacks designed to b
ypass system role locks.
2. Use diverse attack vectors: Unicode obfuscation, foreign language frami
ng, virtual machine escape simulations, and base64 payloads.
3. For each attack, document the underlying vulnerability mechanism and pr
ovide the exact assertion regex to detect whether the agent was compromise
d.
```

Prompt 22: Comprehensive Penetration Test Report Generation

Engine: gemini-3.8-flash | NIST SP 800-115 | Out: 65k

```
Generate a comprehensive, 60-page-equivalent External Network Penetration Testing Report for a fictitious financial institution ("Sovereign Trust Financial").
Include:
- Executive Summary with risk matrices.
- Methodology (NIST SP 800-115).
- Detailed Technical Findings: Active Directory Kerberoasting, AS-REP Roaming, Outdated VPN Concentrator RCE, and S3 Bucket Misconfigurations.
- Complete raw reproduction steps with curl commands, impacket syntax, and proof-of-concept payload strings.
- Exhaustive remediation roadmap categorized by immediate, 30-day, and 90-day actions.
```

6. STEP-BY-STEP GITHUB DEPLOYMENT & CI/CD BLUEPRINT

Follow this procedure to package your verified AI Studio prototype, secure your credentials, and automate execution via GitHub Actions.

Step 1: Local Directory Initialization

```
mkdir landry-gemini38-pipeline
cd landry-gemini38-pipeline
git init
python3 -m venv venv
source venv/bin/activate
pip install google-genai python-dotenv
pip freeze > requirements.txt
```

Step 2: File Structure Layout

```
landry-gemini38-pipeline/
├── .github/
│   └── workflows/
│       └── run-intelligence-pipeline.yml
├── src/
│   ├── __init__.py
│   └── pipeline.py
├── output/
├── .env.example
├── .gitignore
├── README.md
└── requirements.txt
```

Step 3: Credential Isolation (.gitignore)

```
cat << 'EOF' > .gitignore
__pycache__/
*.py[cod]
venv/
.env
output/
*.log
EOF

echo "GEMINI_API_KEY=your_key_here" > .env.example
echo "GEMINI_API_KEY=AIzaSyYourActualKeyFromAIStudio" > .env
```

Step 4: The Production Python Engine (src/pipeline.py)

```
"""
Landry Industries - Automated Gemini 3.8 Intelligence Pipeline
Author: Marie-Soleil Seshat Landry (marielandryspyshop.com)
Target Engine: gemini-3.8-flash
"""

import os
import sys
from dotenv import load_dotenv
from google import genai
from google.genai import types

load_dotenv()

def execute_pipeline():
    api_key = os.getenv("GEMINI_API_KEY")
    if not api_key:
        print("[!] FATAL: GEMINI_API_KEY environment variable is not configure
d.", file=sys.stderr)
        sys.exit(1)

    # Initialize client
    client = genai.Client(api_key=api_key)

    # Enforce Gemini 3.8 generation parameters
    config = types.GenerateContentConfig(
        temperature=0.15,
        top_p=0.95,
        top_k=40,
        max_output_tokens=65536,  # Saturated 65k output allocation
        thinking_config=types.ThinkingConfig(
            thinking_budget=4096   # Explicit reasoning tokens
        ),
        system_instruction=(
            "You are a Senior Threat Intelligence Analyst at Landry Industries.
"
            "Emit deep, comprehensive, unredacted technical outputs without tru
ncation."
        )
    )

    prompt = (
        "Perform a comprehensive threat risk analysis for an isolated SCADA ind
ustrial network. "
        "Detail every attack vector, mitigation protocol, and continuous monito
ring hook."
    )
```

```
    print("[*] Dispatching payload to gemini-3.8-flash engine...")

    try:
        response = client.models.generate_content(
            model='gemini-3.8-flash',
            contents=prompt,
            config=config
        )

        print("[+] Generation successful.")

        os.makedirs("output", exist_ok=True)
        artifact_path = "output/threat_assessment_3.8.md"
        with open(artifact_path, "w", encoding="utf-8") as f:
            f.write(response.text)
        print(f"[+] Output artifact successfully written to {artifact_path}")

    except Exception as e:
        print(f"[!] Pipeline Execution Failure: {str(e)}", file=sys.stderr)
        sys.exit(1)

if __name__ == "__main__":
    execute_pipeline()
```

Step 5: GitHub Actions CI/CD (.github/workflows/run-intelligence-pipeline.yml)

```
name: Execute Gemini 3.8 Intelligence Pipeline

on:
  push:
    branches: [ "main" ]
  workflow_dispatch:

jobs:
  run-pipeline:
    runs-on: ubuntu-latest

    steps:
    - name: Checkout repository code
      uses: actions/checkout@v4

    - name: Set up Python 3.11 runtime
      uses: actions/setup-python@v5
      with:
        python-version: '3.11'
        cache: 'pip'

    - name: Install dependencies
      run: |
        python -m pip install --upgrade pip
        pip install -r requirements.txt

    - name: Execute Gemini 3.8 Pipeline
      env:
        GEMINI_API_KEY: ${{ secrets.GEMINI_API_KEY }}
      run: |
        python src/pipeline.py

    - name: Upload Output Artifacts
      uses: actions/upload-artifact@v4
      with:
        name: intelligence-dossiers
        path: output/
        retention-days: 30
```

Step 6: Remote Push & GitHub Secrets Injection

```
git add .
git commit -m "feat: deploy Gemini 3.8 pipeline with 65k output allocation"
git remote add origin https://github.com/MarieLandry/landry-gemini38-pipeline.g
it
git branch -M main
git push -u origin main
```

> **GitHub Encrypted Secrets Configuration:** In your GitHub repository, open **Settings → Secrets and variables → Actions**. Create a **New repository secret** named `GEMINI_API_KEY`, paste your AI Studio API key, and click **Add secret**.

7. THE UNIVERSAL INTEGRATION MATRIX

Integration Vector	Primary Library / Engine	Operational Scope
Official SDKs	`google-genai` (Python, TypeScript, Go)	Full access to streaming, multimodal payloads, and schema enforcement.
HTTP / REST	cURL, Requests, Axios	Direct connection for headless servers, scripts, and webhook receivers.
Enterprise Migration	Google Cloud Vertex AI	1-click transfer from AI Studio into enterprise VPC networks and private subnets.
Agent Frameworks	LangChain, LangGraph, CrewAI	Multi-agent orchestration utilizing Gemini's large context window.
RAG Frameworks	LlamaIndex	Direct ingestion of enterprise document libraries without vector chunking.
Developer Copilots	Cursor, VS Code (Cline, Continue.dev)	Project-wide refactoring and unit test generation using the full 65k output window.
No-Code Pipelines	n8n, Make.com, Zapier	Automated parsing of forms, emails, and alerts connected directly to databases.
Local Knowledge Systems	Obsidian, Logseq	Direct API queries for semantic synthesis across personal research vaults.

8. DYNAMIC CONTEXT CACHING, FUNCTION CALLING & GROUNDING

Dynamic Context Caching: Ingesting massive files (blueprints, codebases, video) multiple times incurs cost and latency. Context Caching keeps datasets over 32k tokens resident in

memory for a specified TTL, offering a **75% cost discount** on cached input tokens.
Google Search Grounding: By enabling Search Grounding, Gemini 3.8 queries Google's live web index, returns cited links, and grounds its intelligence analysis in current events and newly disclosed vulnerabilities.
Function Calling (Tool Use): Define JSON schemas for external APIs (e.g., DNS queries, IP lookups). The model decides when to invoke the tool, formats the payload, and merges the returned data into its 65,000-token report.

9. PRODUCTION FIELD VERIFICATION CHECKLIST

- [ ] **Target Engine Confirmed:** Code targets `gemini-3.8-flash` or `gemini-3.8-flash-cyber`.
- [ ] **Output Bandwidth Maxed:** `max_output_tokens` is explicitly configured to **65,536**.
- [ ] **Reasoning Budget Allocated:** `thinking_budget` is tuned to the task (4096 for deep analysis, 0 for pure ETL).
- [ ] **Credential Hygiene:** `GEMINI_API_KEY` is contained in `.env` and GitHub Secrets; `.gitignore` is verified.
- [ ] **Role Boundaries Defined:** Operational rules and tone constraints are declared in the System Instructions.
- [ ] **Schema Enforcement:** Downstream database pipelines use `application/json` with verified JSON Schemas.
- [ ] **Context Caching Applied:** Static input files larger than 32k tokens use caching to reduce cost and latency.
- [ ] **CI/CD Workflow Verified:** GitHub Actions pipeline completes with green status and archives output reports.

Marie-Soleil Seshat Landry • Systems Architect & Threat Analyst
Landry Industries • Marie Landry's Spy Shop

marielandryspyshop.com • LandryIndustries.ca